A. Feder Cooper

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I research a variety of topics in reliable, scalable machine learning. My contributions span privacy, security, evaluation of generative-AI systems, MLSys, and measurement validity. I also do work in tech policy and law, and spend a lot of time finding ways to effectively communicate the capabilities and limits of AI/ML to interdisciplinary audiences and the public. My research has received awards at *NeurIPS*, *ICML*, *COLM*, *AAAI*, and *AIES*.

Current Position

Assistant Professor, Department of Computer Science, Yale University
Affiliate Researcher, Stanford University
Faculty Associate at the Berkman Klein Center for Internet & Society at Harvard University

Education

Cornell University
Ph.D., Computer Science 2018 – 2024
M.S., Computer Science 2021

Columbia University
B.A., Computer Science and Archaeology, Phi Beta Kappa, *summa cum laude* 2014

Selected Papers

*Denotes co-first author or equal contribution; †lead author (not specified if same as first author); **Google Scholar**

Subham Sekhar Sahoo, Lingjie Chen, Khiem Pham, Jonathan Geuter, **A. Feder Cooper**, Junlin Chen, Chaitanya Dwivedi, Varad Pimpalkhute, Yash Akhauri, Alexander Moreno, Mikhail Yurochkin, Zhenting Wang, Mostafa Elhoushi, Nolan Dey, Shane Bergsma, Joel Hestness, Hongyi Wang, Hanghang Tong, John Thickstun, Eric Xing, and Zhengzhong Liu. "Unlocking Lossless Speedups in LLMs via Discrete Diffusion." Forthcoming, 2026.

A. Feder Cooper, Marika Swanberg, Jamie Hayes, Lea Duesterwald, Christopher De Sa, Daniel E. Ho, Mark A. Lemley, and Percy Liang. "Extractable Memorization From First Principles." 2026. **arXiv** [link]

Ahmed Ahmed*, **A. Feder Cooper***, Sanmi Koyejo, and Percy Liang. "Extracting books from production language models." 2026. **arXiv** [link]

A. Feder Cooper, Mark A. Lemley, Allison Casasola, Ahmed Ahmed, Aaron Gokaslan, Amy B. Cyphert, Christopher De Sa, Daniel E. Ho, and Percy Liang. "Extracting memorized pieces of (copyrighted) books from open-weight language models." *COLM '26*. **Proceedings** **Oral** (written 2025; ≈0.8% of submitted papers) [link]

A. Feder Cooper, Mark A. Lemley, Christopher De Sa, Lea Duesterwald, Allison Casasola, Jamie Hayes, Katherine Lee, Daniel E. Ho, and Percy Liang. "Estimating near-verbatim extraction risk in language models with decoding-constrained beam search." *COLM '26*. **Proceedings** (29% acceptance; *ICML* Workshop **Oral**) [link]

Meera Desai, Sang T. Truong, Hanna Wallach, Alex Chouldechova, **A. Feder Cooper**, Jean Garcia-Gathright, Daniel E. Ho, Abigail Z. Jacobs, Sanmi Koyejo, Nicholas Pangakis, and Angelina Wang. "What AI Benchmarks Actually Measure: Adapting Convergent and Discriminant Validity to Interrogate Fifty-Six AI Benchmarks." *COLM '26*. **Proceedings** **Oral** (≈0.8% of submitted papers) [link]

Mark A. Lemley* and **A. Feder Cooper***. "Probabilistic 'Copies' in Generative AI Models." *Berkeley Technology Law Journal*. 2026. **Journal** [link]

Miles Brundage, Noemi Dreksler, Aidan Homewood, Sean McGregor, Patricia Paskov, Conrad Stosz, Girish Sastry, **A. Feder Cooper**, George Balston, Steven Adler, Stephen Casper, Markus Anderljung, Grace Werner and others. "Frontier AI Auditing: Toward Rigorous Third-Party Assessment of Safety and Security Practices at Leading AI Companies." 2026. **Whitepaper** [link]

Jamie Hayes*, Ilia Shumailov, Christopher A. Choquette-Choo, Matthew Jagielski, George Kaissis, Milad Nasr, Niloofar Mireshghallah, Meenatchi Sundaram Mutu Selva Annamalai, Igor Shilov, Matthieu Meeus, Yves-Alexandre de Montjoye, Katherine Lee, Franziska Boenisch, Adam Dziedzic, and **A. Feder Cooper***. "Exploring the limits of strong membership inference attacks on large language models." *NeurIPS '25*. Proceedings (24.52% overall acceptance rate) [link]

A. Feder Cooper*†, Christopher A. Choquette-Choo*, Miranda Bogen*, Kevin Klyman*, Matthew Jagielski*, Katja Filippova*, Ken Ziyu Liu*, Alexandra Chouldechova, Jamie Hayes, Yangsibo Huang, Eleni Triantafillou, Peter Kairouz, Nicole Mitchell, Niloofar Mireshghallah, Abigail Z. Jacobs, James Grimmelmann, Vitaly Shmatikov, Christopher De Sa, Ilia Shumailov, Andreas Terzis, Solon Barocas, Jennifer Wortman Vaughan, danah boyd, Yejin Choi, Sanmi Koyejo, Fernando Delgado, Percy Liang, Daniel E. Ho, Pamela Samuelson, Miles Brundage, David Bau, Seth Neel, Hanna Wallach, Amy B. Cyphert, Mark A. Lemley, Nicolas Papernot, and Katherine Lee*†. "Machine Unlearning Doesn't Do What You Think: Lessons for Generative AI Policy and Research." *NeurIPS '25*, *Position Track*. Proceedings Oral (≈0.4% of submitted papers). [link]

Jamie Hayes*, Marika Swanberg, Harsh Chaudhari, Itay Yona, Ilia Shumailov, Milad Nasr, Christopher A. Choquette-Choo, Katherine Lee, and **A. Feder Cooper***. "Measuring memorization in language models via probabilistic extraction." *NAACL '25*. Proceedings (≈22% overall acceptance rate) [link]

A. Feder Cooper* and James Grimmelmann*. "The Files are in the Computer: On Copyright, Memorization, and Generative AI." *Chicago-Kent Law Review*, Vol. 100, 2025. Journal (written 2024) [link]

Katherine Lee*, **A. Feder Cooper***, and James Grimmelmann*. "Talkin' 'Bout AI Generation: Copyright and the Generative-AI Supply Chain." *Journal of the Copyright Society*, Volume 72, 2025. Journal (written 2023) [link]

Nicholas Carlini, Daniel Paleka, Krishnamurthy Dj Dvijotham, Thomas Steinke, Jonathan Hayase, **A. Feder Cooper**, Katherine Lee, Matthew Jagielski, Milad Nasr, Arthur Conmy, Eric Wallace, David Rolnick, and Florian Tramèr. "Stealing Part of a Production Language Model." *ICML '24*. Proceedings Best Paper (∼0.1% of submitted papers) [link]

A. Feder Cooper, Katherine Lee, Madiha Zahrah Choksi, Solon Barocas, Christopher De Sa, James Grimmelmann, Jon Kleinberg, Siddhartha Sen, and Baobao Zhang. "Arbitrariness and Social Prediction: The Confounding Role of Variance in Fair Classification." *AAAI '24*. Proceedings Best Student Paper Honorable Mention (∼0.3% of submitted papers; written 2023) [link]

A. Feder Cooper*†, Wentao Guo*, Khiem Pham*, Tiancheng Yuan, Charlie F. Ruan, Yucheng Lu, and Christopher De Sa†. "Coordinating Distributed Example Orders for Provably Accelerated Training." *Conference on Neural Information Processing Systems 36 (NeurIPS '23)*. Proceedings (26% overall acceptance rate) [link]

Milad Nasr*, Nicholas Carlini*, Jonathan Hayase, Matthew Jagielski, **A. Feder Cooper**, Daphne Ippolito, Christopher A. Choquette-Choo, Eric Wallace, Florian Tramèr, and Katherine Lee†. "Scalable Extraction of Training Data from (Production) Language Models." 2023. arXiv [link]

A. Feder Cooper*†, Katherine Lee*, James Grimmelmann*, Daphne Ippolito*, Christopher Callison-Burch, Christopher A. Choquette-Choo, Niloofar Mireshghallah, Miles Brundage, David Mimno, Madiha Zahrah Choksi, Jack M. Balkin, Nicholas Carlini, Christopher De Sa, Jonathan Frankle, Deep Ganguli, Bryant Gipson, Andres Guadamuz, Swee Leng Harris, Abigail Z. Jacobs, Elizabeth Joh, Gautam Kamath, Mark Lemley, Cass Matthews, Christine McLeavey, Corynne McSherry, Milad Nasr, Paul Ohm, Adam Roberts, Tom Rubin, Pamela Samuelson, Ludwig Schubert, Kristen Vaccaro, Luis Villa, Felix Wu, and Elana Zeide. "Report of the 1st Workshop on Generative AI and Law." 2023. Whitepaper [link]

A. Feder Cooper, Jonathan Frankle, and Christopher De Sa. "Non-Determinism and the Lawlessness of Machine Learning Code." *CSLAW '22*. Proceedings [link]

A. Feder Cooper*†, Emanuel Moss*, Benjamin Laufer, and Helen Nissenbaum†. "Accountability in an Algorithmic Society: Relationality, Responsibility, and Robustness in Machine Learning." *FAccT '22*. Proceedings (25% overall acceptance rate) [link]

A. Feder Cooper, Yucheng Lu, Jessica Zosa Forde, and Christopher De Sa. "Hyperparameter Optimization Is Deceiving Us, and How to Stop It." *NeurIPS '21*. Proceedings (<26% overall acceptance rate) [link]

Ruqi Zhang*, **A. Feder Cooper***, and Christopher De Sa. "Asymptotically Optimal Exact Minibatch Metropolis-Hastings." *NeurIPS '20*. **Proceedings** **Spotlight** ($<3\%$ of submitted papers) [**link**]

Selected Honors, Grants, and Awards

Stanford HAI Seed Grant ($75,000), "Assessing Copyright Risks of Training Data 'Memorization' and 'Extraction' for Open-Weight Language Models" (2026 – 2027)

Top Area Chair, *NeurIPS* (2026)

Meta Ph.D. Fellowship Finalist (2023) (short-listed out of 3200 candidates)

Outstanding *ICML* Reviewer ($\leq 10\%$ of reviewers) (2022)

Rising Star in EECS, MIT (2021)

Fellowship Finalist, OpenPhil AI Ph.D. Fellowship (2021)

Selected Conference, Workshop, and Tutorial Organization

Knight Foundation convening on generative AI and copyright. November 2026. Co-organized with James Grimmelmann.

Data Privacy, Memorization, & Legal Implications in Generative AI: A Practical Guide. *NeurIPS '25* Tutorial, December 2, 2025 (San Diego, CA). Co-organized with Pratyush Maini and Joe Gratz.

2nd Workshop on Generative AI and Law (GenLaw '24). Held at *ICML 2024* in Vienna, Austria. Lead co-organizer with Katherine Lee. [**link**]

Evaluating Generative AI Systems: the Good, the Bad, and the Hype (GenLaw DC '24). Lead co-organizer. Sponsored by the K&L Gates initiative in Ethics and Computational Technologies at Carnegie Mellon University (CMU), and organized by the GenLaw Center (A. Feder Cooper, Katherine Lee, James Grimmelmann), Carnegie Mellon's K&L Gates Initiative (Hoda Heidari), the Georgetown Law Center (Paul Ohm), and the Center for Democracy and Technology (Alexandra Reeve Givens, Miranda Bogen). [**link**]

1st Workshop on Generative AI and Law (GenLaw '23). Held at *ICML 2023* in Honolulu, Hawai'i, 26% acceptance rate. Lead co-organizer with Katherine Lee; co-organized with Dr. Niloofar Mireshghallah, Madiha Zahrah Choksi, Prof. James Grimmelmann, Prof. David Mimno, and Dr. Deep Ganguli [**link**]

Selected Talks and Panels

"Closing the book on memorization, language models, and copyright," Yale Institute for the Foundations of Data Science, September 30, 2026 (New Haven, CT).

Invited participant, Ronald M. Whyte Intellectual Property Seminar, September 1–4, 2026 (Berkeley, CA).

"Extractable Memorization from First Principles." **2nd Workshop on the Impact of Memorization on Trustworthy Foundation Models**, *International Conference on Machine Learning*, July 2026 (Seoul, Korea).

Invited speaker, 29th Annual BTLJ–BCLT Spring Symposium: Origins, Evolution, and Possible Futures of the 1976 Copyright Act, April 16–17, 2026 (Berkeley, CA).

Invited participant, Symposium on AI and Law, Harvard Law School, October 24, 2025 (Cambridge, MA).

"What Copyright Can Learn from Memorization Measurements of Language Models."

Invited talk, American National Standards Institute (ANSI), May 11, 2026 (Virtual).

Invited speaker, Workshop & Lecture Series on the Law & Economics of Innovation, ETH Zürich, October 28–29, 2025 (Zürich, Switzerland).

Invited speaker, Information Technology and Innovation Foundation, August 2025 (Virtual).

Invited speaker, **Workshop on the Impact of Memorization on Trustworthy Foundation Models**, *International Conference on Machine Learning*, July 2025 (Vancouver, Canada).

Invited panelist, **Workshop on Machine Unlearning for Generative AI**, *International Conference on Machine Learning*, July 2025 (Vancouver, Canada).

AI Governance Tutorial (with Katherine Lee, 28th Annual BTLJ-BCLT Spring Symposium: AI Governance at the Crossroads, February 27, 2025 (Berkeley, CA).

"Machine Unlearning Doesn't Do What You Think: Lessons for Generative AI Policy, Research, and Practice."

Future of Privacy Forum, Technologist Roundtable for Policymakers, July 2025 (Virtual).

Invited talk, NYU Law Innovation Policy Colloquium, April 3, 2025 (New York, NY).

Microsoft Research, January 2025 (New York, NY).

Future of Privacy Forum, January 2025 (Virtual – Washington, DC).

Selected Service

Area Chair for *COLM* (2025-2026), *ICML* (2026), *NeurIPS* (2026), *ICML* (2027); Action Editor for *TMLR* (2026 –)

Publicity co-chair for *CSLAW* (2025-2026)

Reviewer / Program Committee for *NeurIPS* (2020-2024), *ICML* (2021-2023, 2025), *ICLR* (2022-2023, 2025-2026), *TMLR* (2022), *CSLAW* (2025–2027), *ACM CACM* (2025-2026), *COLM* (2024)